



**Revere Plastics
Systems, LLC.**

Gavin Bahner
Summer Internship '25

Main Projects

Process Verification

- Corrected
- Optimized
- Filling out Daily ~ 24 Presses

test-MSU-12345-4-999.xlsx.xlsm - ThisWorkbook (Code)

Workbook

Before Save

```

If ws.Visible = xlSheetVisible Then
    sheetInfo = "Page: " & ws.Name
    Dim localMessage As String
    localMessage = ""
    For j = 7 To 60 Step 2
        If ws.Cells(30, j).Value = "" Then 'Uses date as marker for case of skipped/marked days
            j = j - 2
            Exit For
        End If
    Next j
    For i = 2 To 30 Step 1 'Scan all formula cells
        newValue = LCase(CStr(ws.Cells(i, j - 1).Value)) 'converts cell value to string and makes lower
        Debug.Print (newValue)
        Dim desc As String, expectedVal As String, actualVal As String
        Dim nameVal As String, infoVal As String
        If newValue = "false" Then
            desc = ws.Cells(i, 1).Value ' Column A
            expectedVal = ws.Cells(i, 2).Value ' Column B
            actualVal = CStr(ws.Cells(i, j - 1).Value)
            infoVal = ws.Cells(i, j).Value ' Info next to FALSE

            If Not changeDatabase.exists(ws.Name & ws.Cells(i, j).Address) Then
                'Debug.Print (ws.Cells(i, j).Address)
                foundNewFalse = True
                changeDatabase(ws.Name & ws.Cells(i, j).Address) = newValue 'assign to general database
            End If
            'nameVal = ws.Cells(28, cell.Column + 1).Value 'Tech initials in row 28

            ' Adjust actual value display
            If IsError(ws.Cells(i, j - 1).Value) Then
                actualVal = "Error"
            ElseIf VarType(ws.Cells(i, j - 1).Value) = vbBoolean Then
                If IsEmpty(ws.Cells(i, 2)) Then

```

	A	B	C	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
1	VA08 Bal Upper	MASTER SETUP	TOL +/-																	
2	Shot Size	3.50	0.50	9.5	9.5	9.5	9.5	9.5	9.5	9.5	9.5	9.5	9.5	9.5	9.5	9.5	9.5	9.5	9.5	9.5
3	Inj PSI	1120	200	1103	1049	1151	1149	1085	1089	1084	1063	1063	1063	1063	1063	1063	1063	1063	1063	1063
4	Inj Time	3.60	1.00	3.6	3.366	3.396	3.128	3.302	3.292	3.226	3.202	3.202	3.202	3.202	3.202	3.202	3.202	3.202	3.202	3.202
5	Xfer POS.	2.00	0.50	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2
6	Pack PSI	850	200	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850
7	Hold PSI	850	200	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850
8	Pack Time	1.00	0.50	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
9	Hold Time	6.00	1.00	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6
10	Cushion	0.37	0.50	0.38	0.36	0.35	0.37	0.35	0.35	0.35	0.35	0.35	0.35	0.35	0.35	0.35	0.35	0.35	0.35	0.35
11	Back Pressure	50	50	50	50	50	50	50	50	50	50	50	50	50	50	50	50	50	50	50
12	Recovery Time	13.60	5.00	13.66	13.908	13.772	13.392	14.126	13.688	13.428	14.27	14.27	14.27	14.27	14.27	14.27	14.27	14.27	14.27	14.27
13	RPM	100	25	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100
14	D-Comp.	0.50	0.50	0.5	0.25	0.25	0.25	0.25	0.25	0.25	0.25	0.25	0.25	0.25	0.25	0.25	0.25	0.25	0.25	0.25
15	Heats																			
16	Nozzle	400	50	400	400	400	400	400	400	400	400	400	400	400	400	400	400	400	400	400
17	Barrel 1	400	50	400	400	400	400	400	400	400	400	400	400	400	400	400	400	400	400	400
18	Barrel 2	400	50	400	400	400	400	400	400	400	400	400	400	400	400	400	400	400	400	400
19	Barrel 3	350	50	390	390	390	390	390	390	390	390	390	390	390	390	390	390	390	390	390
20	Barrel 4																			
21	Barrel 5																			
22	Barrel 6																			
23	Cramp Open Position	33.85	0.10	33.85	33.85	33.85	33.85	33.85	33.85	33.85	33.85	33.85	33.85	33.85	33.85	33.85	33.85	33.85	33.85	33.85
24	Stationary Half Water	CHILL																		
25	Movable Half Water	CHILL																		
26	Cooling Time	26.00	2.00	26	26	26	26	30	30	30	31	31	31	31	31	31	31	31	31	31
27	Cycle Time	50.00	2.00	49.89	48.76	48.984	48.624	52.82	52.842	52.736	53.732	53.732	53.732	53.732	53.732	53.732	53.732	53.732	53.732	53.732
28	Tech Initials			GB	GB	GB	GB	GB	GB	GB	GB	GB	GB	GB	GB	GB	GB	GB	GB	GB
29	Shift			1st	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
30	Date			6/12/2025	6/13/2025	6/16/2025	6/17/2025	6/18/2025	6/19/2025	6/20/2025	6/23/2025	6/23/2025	6/23/2025	6/23/2025	6/23/2025	6/23/2025	6/23/2025	6/23/2025	6/23/2025	6/23/2025
31																				
32	Press #		100																	
33	Mold #		4076-2																	
34	MSU #		MSU-4076-2-100																	
35	Date on Setup		6/9/2025																	
36																				

< >

Inputs-Outputs

Hot Runners

Valve Gates

+

Lingering Questions

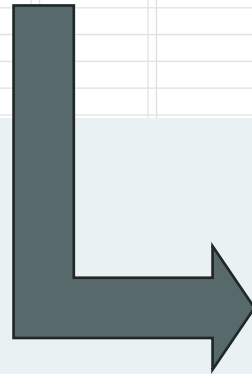
- How to integrate this into the current workflow?
- Is this the best solution?

Workflow Optimization

- UI form to help tech speed up process

Press #	999						
Mold #	12345-4						
Shot #							
Injection Setup							

Add New Verification



Process Verification Editor

Fill Time

Recovery Time

Cycle Time

Cushion

Injection Pressure



Barrel Temperature

Nozzle:

Front:

Back:

Clamp Open Position

Screw Speed

Inject Time

Back Pressure

Cooling Time

Press: 999

Mold: 12345-4

Part: Sample Name

Initials:

Shift:

Date: DD/MM/YYYY

Pack Time

Pack Pressure

Hold Pressure

Transfer Position

Shot Size

Decompression

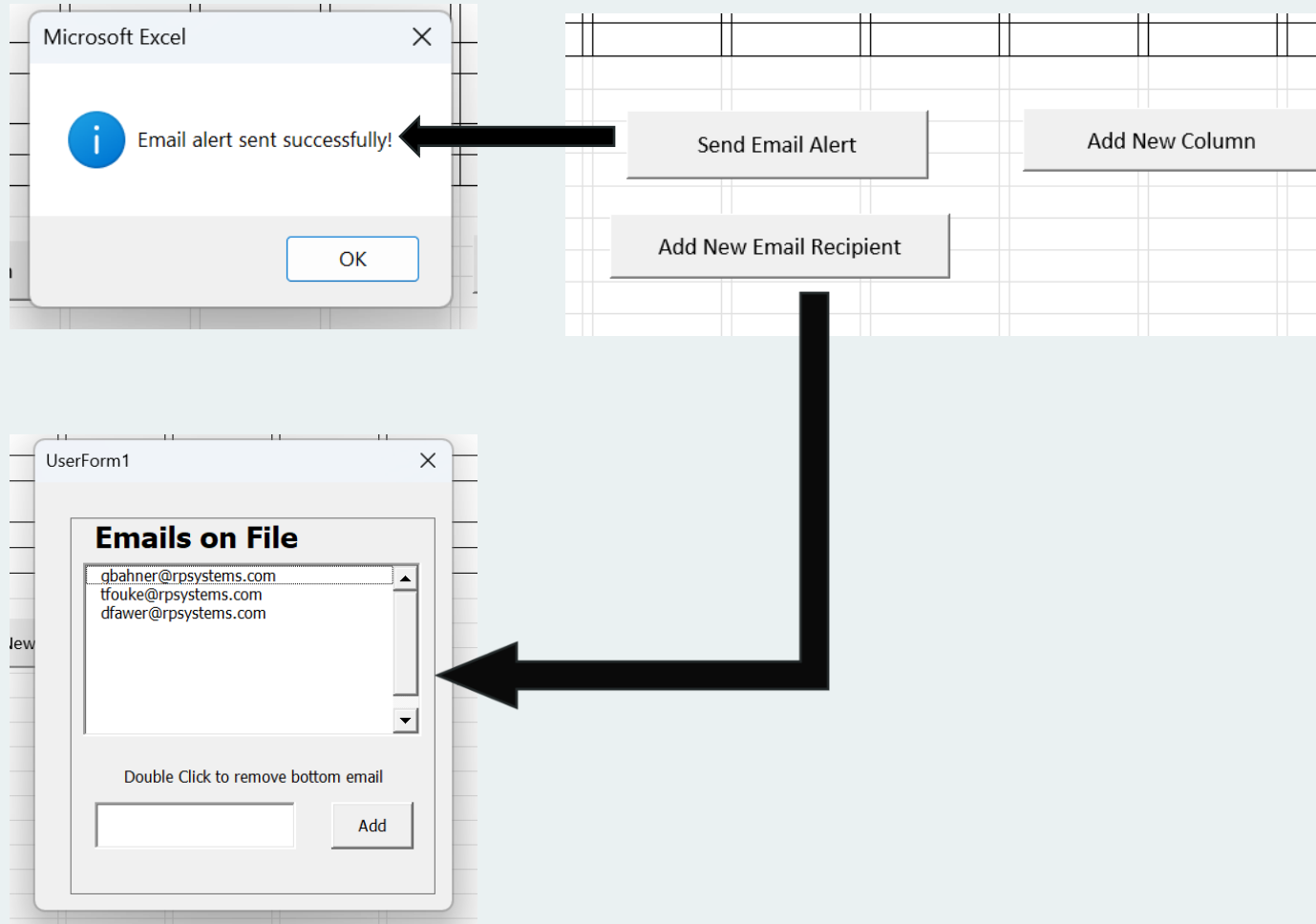
Transfer Data to Excel Sheet

Save Excel Sheet

Exit

Workflow Integration

- Added sheet features to ensure seamless long-term operation



Procces Altertation Alert - Press: 181



Bahner, Gavin

To: ✓ Bahner, Gavin



9:06 AM

FILE: PROCESS VERIFICATION-MSU-EOS00020-2-181.xlsm

Press: 181

Page: Inputs-Outputs

- Parameter: Shot Size (Cell \$M\$2)

New Value: 57

Set Value: 55

Issue: Outside of Tolerance

Workflow Integration

- Added ability to toggle between states for increased readability

A	B	C	F	J	L	N	P	R	T	V	X	Z	AA	AC	AE	AG	AI
Bipolar Stiffener OM	MASTER SETUP [mm, s, F, %]	TOL +/-	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Shot Size	55.00	0.50	55	55	55	57	55	55	55	55	55	55	55	55	55	55	55
Inj PSI	300	75	303	320	311	314	311	311	311	311	311	311	311	311	311	311	311
Inj Time	5.00	0.50	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5
Fill Time	4.10	0.50	3.82	3.85	3.86	3.93	3.86	3.86	3.86	3.86	3.86	3.86	3.86	3.86	3.86	3.86	3.86
Xfer POS.	22.00	3.00	25	25	25	25	25	25	25	25	25	25	25	25	25	25	25
Pack PSI	5%	5%	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05
Hold PSI	5%	5%	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05
Pack Time	1.00	0.25	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Hold Time	0.00	0.50	0.00	0.50	0.00	0.50	0.00	0.50	0.00	0.50	0.00	0.50	0.00	0.50	0.00	0.50	0.00
Cushion	20.00	2.00	21.6	21.93	21.66	21.78	21.66	21.66	21.66	25	21.66	22	21.66	17	21.66	23	
Back Pressure	15%	5%	15%	15%	15%	15%	15%	15%	15%	15%	15%	15%	15%	15%	15%	15%	15%
Recovery Time	3.00	1.00	3.01	3.04	3.05	3.04	3.05	3.05	3.05	3.05	3.05	3.05	3.05	3.05	3.05	3.05	3.05
RPM	35%	5%	0.35	0.35	0.35	0.35	0.35	0.35	0.35	0.35	0.35	0.35	0.35	0.35	0.35	0.35	0.35
D-Comp.	5.00	5.00	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5
Heats																	
Nozzle	400	25	400	400	400	400	400	400	400	400	400	400	400	400	400	400	400
Barrel 1	385	25	385	385	385	385	385	385	385	385	385	385	385	385	385	385	385
Barrel 2	385	25	385	385	385	385	385	385	385	385	385	385	385	385	385	385	385
Barrel 3	380	25	380	380	380	380	380	380	380	380	380	380	380	380	380	380	380
Barrel 4	375	25	375	375	375	375	375	375	375	375	375	375	375	375	375	375	375
Barrel 5																	
Barrel 6																	
Clamp Open Position	695.00	10.00	695	695	695	695	695	695	695	695	695	695	695	695	695	695	695
Stationary Half Water	85	15															
Movable Half Water	85	15															
Cooling Time	5.00	0.50	5	5	5	5	5	7	5	5	5	5	9	5	5	5	5
Cycle Time	33.00	5.00	33.19	33.22	33.01	33.08	33.01	33.01	33.01	33.01	33.01	33.01	33.01	33.01	33.01	33.01	33.01
Tech Initials			GB	GB	GB	GB	GB	GB	GB	GB	GB	GB	GB	GB	GB	GB	GB
Shift			1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Date			7/24/2025	7/28/2025	8/7/2025	8/8/2025	8/9/2025	8/11/2025	8/12/2025	8/13/2025	8/14/2025	8/15/2025	8/17/2025	8/18/2025	8/19/2025	8/20/2025	



A	B	C	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	AA	AC	AE	AG	AI	
Bipolar Stiffener OM	MASTER SETUP [mm, s, F, %]	TOL +/-	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15												
Shot Size	55.00	0.50	55	55	55	57	55	55	55	55	55	55	55	55	55	55	55	55	55	55	55	55	55	55	55	55	55		
Inj PSI	300	75	303	320	311	314	311	311	311	311	311	311	311	311	311	311	311	311	311	311	311	311	311	311	311	311	311		
Inj Time	5.00	0.50	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5		
Fill Time	4.10	0.50	3.82	3.85	3.86	3.93	3.86	3.86	3.86	3.86	3.86	3.86	3.86	3.86	3.86	3.86	3.86	3.86	3.86	3.86	3.86	3.86	3.86	3.86	3.86	3.86	3.86		
Xfer POS.	22.00	3.00	25	25	25	25	25	25	25	25	25	25	25	25	25	25	25	25	25	25	25	25	25	25	25	25	25		
Pack PSI	5%	5%	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05		
Hold PSI	5%	5%	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05		
Pack Time	1.00	0.25	1	1	1	1	1	1	2	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1		
Hold Time	0.00	0.50																											
Cushion	20.00	2.00	21.6	21.93	21.66	21.78	21.66	21.66	21.66	25	21.66	22	21.66	17	21.66	23													
Back Pressure	15%	5%	15%	15%	15%	15%	15%	15%	15%	15%	15%	15%	15%	15%	15%	15%	15%	15%	15%	15%	15%	15%	15%	15%	15%	15%	15%		
Recovery Time	3.00	1.00	3.01	3.04	3.05	3.04	3.05	3.05	3.05	3.05	3.05	3.05	3.05	3.05	3.05	3.05	3.05	3.05	3.05	3.05	3.05	3.05	3.05	3.05	3.05	3.05	3.05		
RPM	35%	5%	0.35	0.35	0.35	0.35	0.35	0.35	0.35	0.35	0.35	0.35	0.35	0.35	0.35	0.35	0.35	0.35	0.35	0.35	0.35	0.35	0.35	0.35	0.35	0.35	0.35		
D-Comp.	5.00	5.00	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5		
Heats																													
Nozzle	400	25	400	400	400	400	400	400	400	400	400	400	400	400	400	400	400	400	400	400	400	400	400	400	400	400	400		
Barrel 1	385	25	385	385	385	385	385	385	385	385	385	385	385	385	385	385	385	385	385	385	385	385	385	385	385	385	385		
Barrel 2	385	25	385	385	385	385	385	385	385	385	385	385	385	385	385	385	385	385	385	385	385	385	385	385	385	385	385		
Barrel 3	380	25	380	380	380	380	380	380	380	380	380	380	380	380	380	380	380	380	380	380	380	380	380	380	380	380	380		
Barrel 4	375	25	375	375	375	375	375	375	375	375	375	375	375	375	375	375	375	375	375	375	375	375	375	375	375	375	375		
Barrel 5																													
Barrel 6																													
Clamp Open Position	695.00	10.00	695	695	695	695	695	695	695	695	695	695	695	695	695	695	695	695	695	695	695	695	695	695	695	695	695		
Stationary Half Water	85	15																											
Movable Half Water	85	15																											
Cooling Time	5.00	0.50	5	5	5	5	5	7	5	5	5	5	9	5	5	5	5	5	5	5	5	5	5	5	5	5	5		
Cycle Time	33.00	5.00	33.19	33.22	33.01	33.08	33.01	33.01	33.01	33.01	33.01	33.01	33.01	33.01	33.01	33.01	33.01	33.01	33.01	33.01	33.01	33.01	33.01	33.01	33.01	33.01	33.01		
Tech Initials			GB	GB	GB	GB	GB	GB	GB	GB	GB	GB	GB	GB	GB	GB	GB	GB	GB	GB	GB	GB	GB	GB	GB	GB	GB		
Shift			1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1		
Date			7/24/2025	7/28/2025	8/7/2025	8/8/2025	8/9/2025	8/11/2025	8/12/2025	8/13/2025	8/14/2025	8/15/2025	8/17/2025	8/18/2025	8/19/2025	8/20/2025													
Inputs-Outputs			Hot Runners			Valve Gates			+			:																	

Inputs-Outputs

Hot Runners

Valve Gates

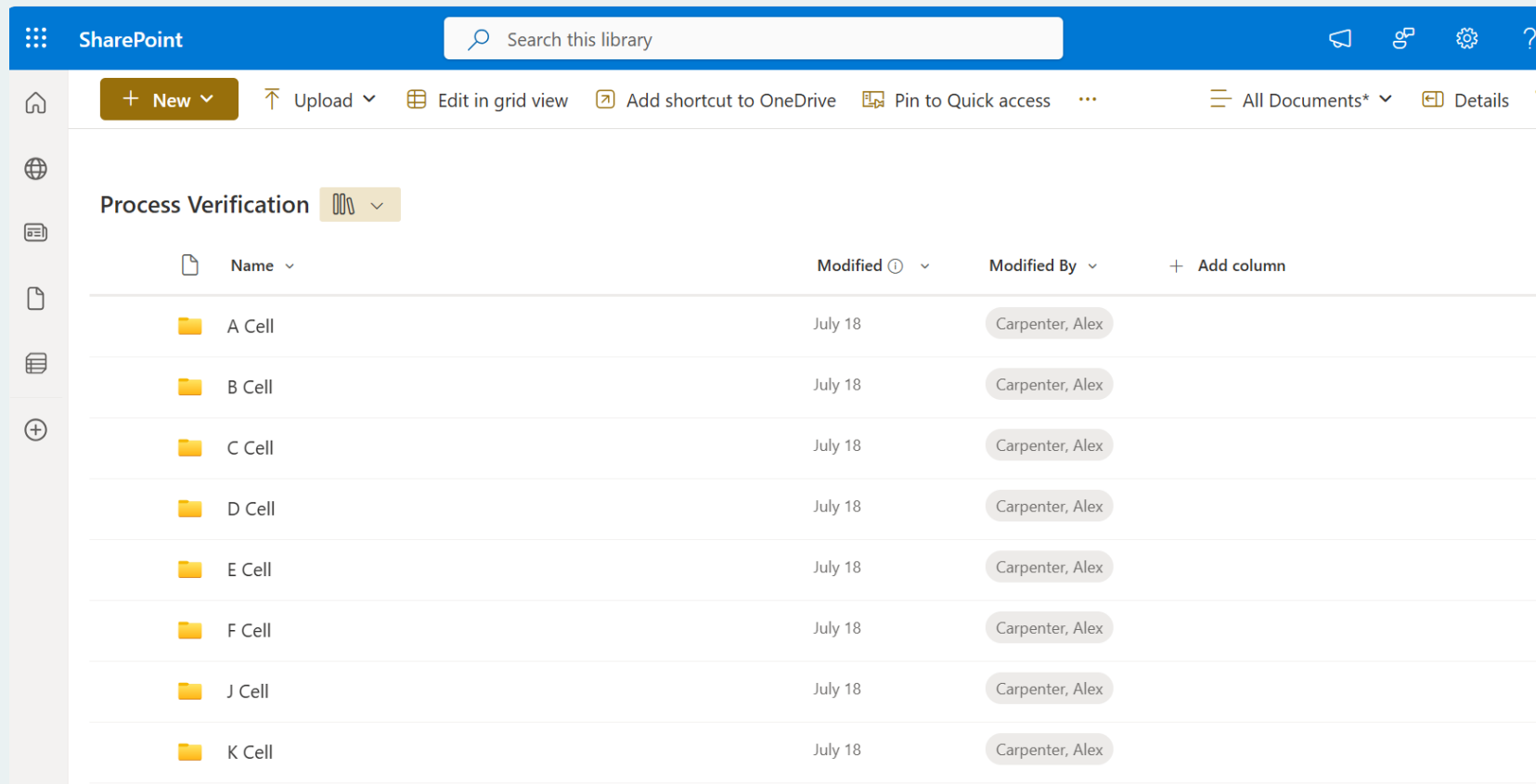
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SharePoint Transition

- Worked with IT Alex to move: P Drive → SharePoint
- Multiple people can access at a single time

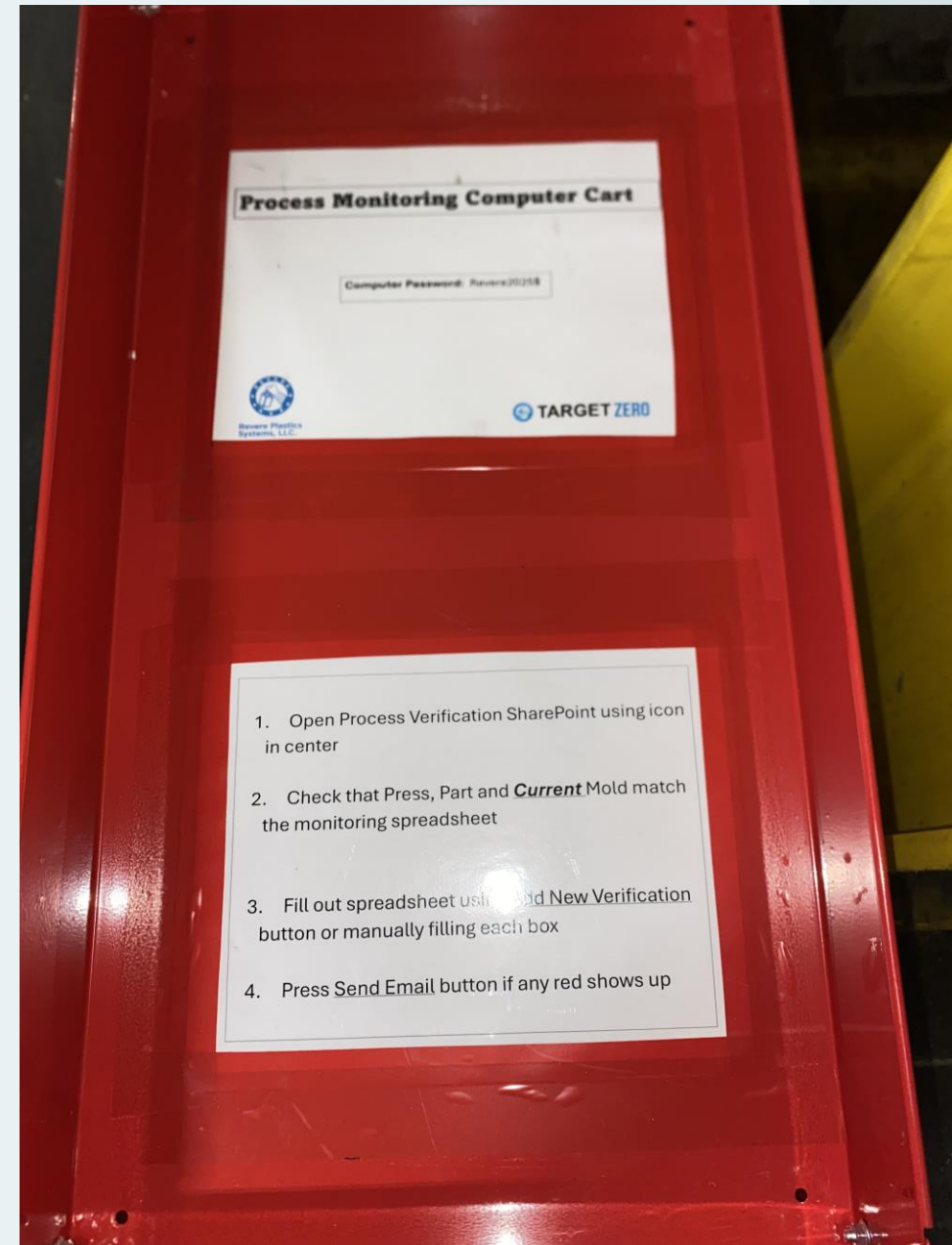


The screenshot displays a SharePoint library interface. At the top, there is a blue header bar with the 'SharePoint' logo, a search bar labeled 'Search this library', and several utility icons. Below the header, a secondary bar contains action buttons: '+ New', 'Upload', 'Edit in grid view', 'Add shortcut to OneDrive', 'Pin to Quick access', and a dropdown menu currently set to 'All Documents*'. The main content area is titled 'Process Verification' and features a table with the following columns: 'Name', 'Modified', 'Modified By', and an '+ Add column' button. The table lists eight folders, each named 'X Cell' (where X is A through K), all of which were modified on 'July 18' by 'Carpenter, Alex'.

Name	Modified	Modified By	+ Add column
A Cell	July 18	Carpenter, Alex	
B Cell	July 18	Carpenter, Alex	
C Cell	July 18	Carpenter, Alex	
D Cell	July 18	Carpenter, Alex	
E Cell	July 18	Carpenter, Alex	
F Cell	July 18	Carpenter, Alex	
J Cell	July 18	Carpenter, Alex	
K Cell	July 18	Carpenter, Alex	

Workflow Integration

- Setup laptop exclusively for process verification
- Acquired cart for storage/transportation
- Trained process techs



Created Priority Sheet

- Used for systematic implementation

1	2	A	B	C	D	E	F	G	H	I
	1	Cell	Press	Current Part	Current Part Name	Current Mold	Monitoring Setup (Y/N)	Monitoring Setup Date		
	2	A Cell								
	11	B Cell								
	12	B	85		240 BR Upper		N			
	13	B	86		240 BR Lower		N			
	14	B	99	W159139	VA08 BR Lower	4077-2	Y			
	15	B	100	W159138	VA08 BR Upper	4076-1	Y			Monitoring Differe
	16	B	127	W11396262	Bezel - Faucet White	4114	Y			
	17	B	164	W159180	2.0 Vmax BR Lower	4102-3	Y			
	18	B	169	W159275/W159267	18/20" Air Tower	4066	Y			
	19	B	1301	8534952	Accumulator Cover	3128-3	N			
	20	B	1403	W159180	2.0 Vmax BR Upper	4103-3	Y			Monitoring Differe
	21	B	4119	W158946	Alpha Lint Screen Door	3663-2	N			
	22	B	8108	W159418	Rail Switch Activator	51886	N			
	23	C Cell								
	33	D Cell								
	42	E Cell								
	52	F Cell								
	62	J Cell								
	73	K Cell								
	90	H Cell								
	101	O Cell								
	118	S Cell								
	137									
	138	+ Buttons to Expand				Total Presses:	124			
	139	1 Button to Collapse All				Total Presses Monitoring:	22			
	140	2 Button to Expand All				Monitoring Percentage:	17.74%			
	141									
	142									
	143									
	144	Monitoring Priority List:								
	145	1.								
	146	2.								
	147									

Lingering Questions

- How to integrate this into the current workflow?
- Is this the best solution?

Best Solution?

- Inconsistency of Excel, slower with more data
- I am a short-term solution AND Process Techs are busy
- Process Verification gives single snapshot of the machine only at that exact moment of that shift

First Question:

Best Solution?



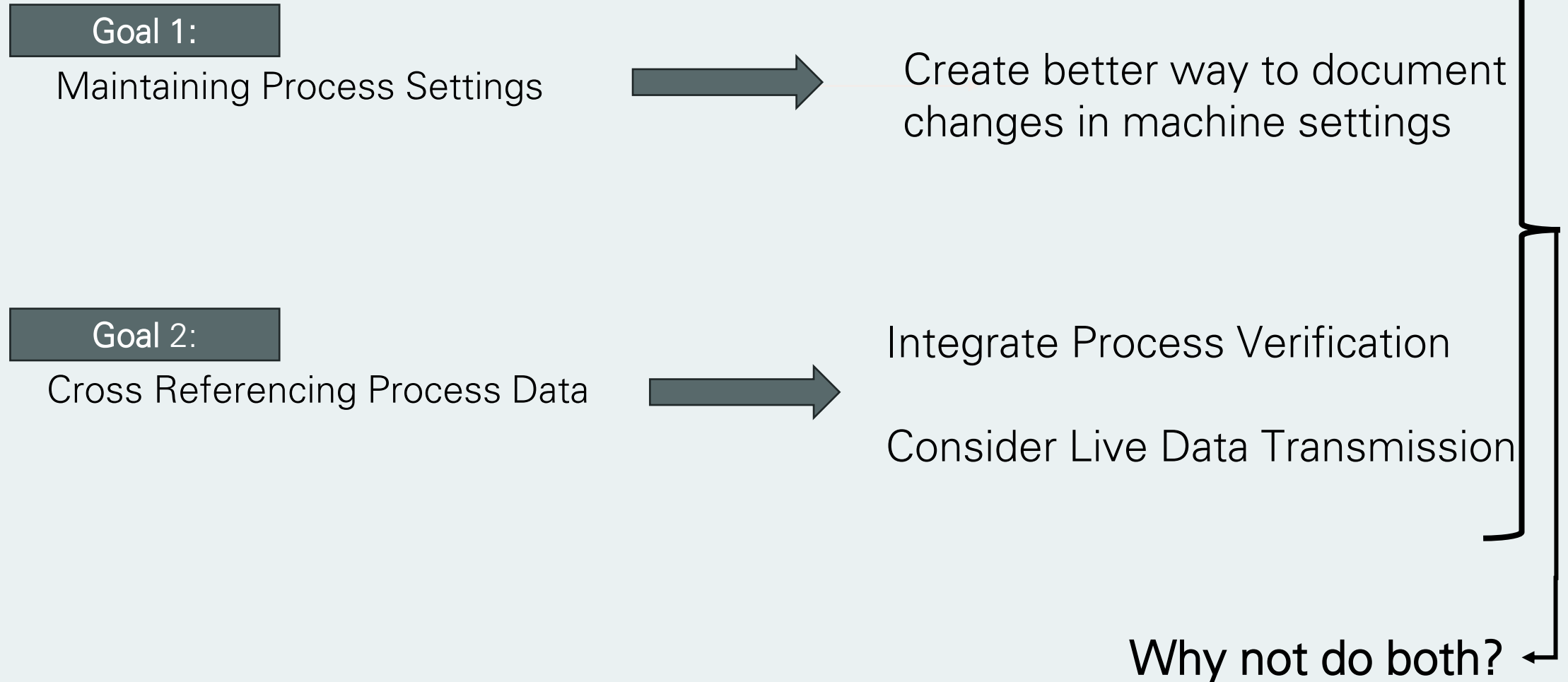
Better First Question:

What is the Real Problem?

My Take on the Problem

- People (process techs, line workers, ...) making undocumented changes on mold machine settings
- Gap in understanding between actual process settings, process output, and output part quality

Best Solution Depends on the Goal



Red X Processes

- Goal: Ensure part testing equipment is working correctly
- Focus: Plug Leak Testers for all Balance Rings

First Iteration: Known Bad Part

- Completed: Fusion Red X Process
- Conflicts of interest, high level risk of breaking carousels
- Interrupts workflow: must clear machine out

Red X documentation hanging at the Air Leak Tester door

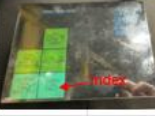
Step 1 Review the Red X sheet Log looking for the last test validation (make sure it was done at the beginning of the last shift).
Record the New Test Validation in the Red X sheet log (Quality Auditor will check this step is done during the shift quality checks for the part)

Quality - Red X data Collection Sheet				TARGET ZERO	
Part No.	W1058655	Container #	11111111	Test Date	1/15/2020
Part Name	Balance Ring Arm	Leak Test Validation	Once at the start of the shift	Test	1
Record Test Results/Defects					
Date	Shift	Time	Name	Defective part Result	Comments
				Fail	Accept
				Fail	Accept

Step 2 a) Using the HMI, place the carousel in manual mode (use substeps as guide) - Make sure to home all positions
HMI Step-by-step: Stop Auto Cycle > Mode Reset > Enable Manual Mode > *****Home all Partitions*****
b) Place Red X part in new part slot of carousel
c) Index the carousel until Red X part is under the last tester (plug leak tester), removing/scraping any parts that exit the carousel
HMI Step-by-step: Gate Screen Selector > Dial Index to Index Dial Table
c1 - Indexing: Press and hold *Index Dial Table* button - if "Dial Not in Position", hold button until in position
c2 - Final Position: Reference the image for the correct location of the Red X part
d) Run the leak test using result displayed by Cincinnati test system unit
HMI Step-by-step: Gate Screen Selector > Leak Test > Lower Leak Test Slide > Start Leak Test Chamber 1
- Fail: Tester is working as it is supposed to be
- Accept: Notify supervisor/Quality Team Imm
If tester accept bad parts. Hold any inventory produced since last good Tester Validation.
e) Swap the Red X part for a good part and run the Test again, close the door and start the cycle.

a)  

b) 

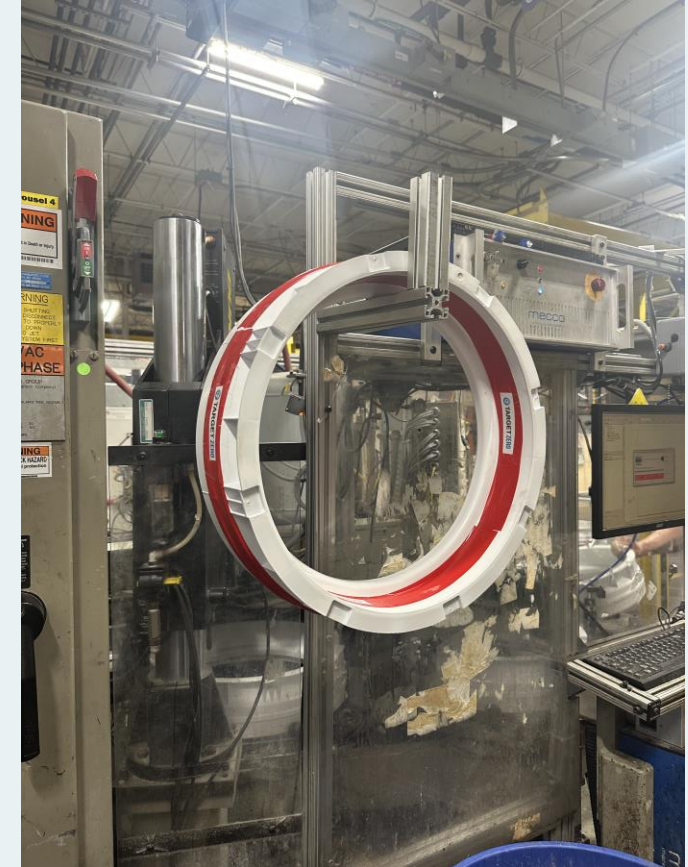
c1) 

c2) 

d) Results:  

Red X part 

PI-8566198C2 Red X sheet Log +

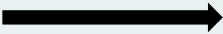


Second Iteration: Known Bad Plug

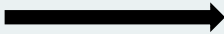
- Use of a visually marked, known bad plug
- Created process for VMAX 2.0 BR – exact same implementations for all BR
- Issue with this method: creating the bad plugs

Options From Here

Option 1:

ME re-takeover of Red X processes  Acquire/create calibrated known bad parts, utilize relatively complex process, rely on ME's for execution

Option 2:

Pursue process, technology to create known bad plugs (with acceptable leak rates)  Easy implementation, simple training, Greater likelihood of longevity

Organizing New PM Scheduling Interface

Menu

Dashboard

Calendar

Work Orders

Assets

Parts & Purchase Orders

Safety Programs

Admin

Revere Plastics Systems, LLC.

PM SCHEDULES

3 Resource Center

Help

Profile

+ New

Delete

More ...

Print/Report

Columns

View

No View - Default

Advanced Filters

Location

Category

Site: Clyde

[All]

A-CELL

B-CELL

C-CELL

D-CELL

E-CELL

F-CELL

H-CELL

J-CELL

K-CELL

O-CELL

S-CELL

☐	Title	Asset	Location	Assigned
☐	Press 99 Semi-annual PM	Press #99	B-CELL	Mathew
☐	Press 99 Quarterly PM	Press #99	B-CELL	Mathew
☐	Press 99 Annual PM	Press #99	B-CELL	Mathew
☐	Press 96 Semi-annual PM	Press #96	C-CELL	Mathew
☐	Press 96 Quarterly PM	Press #96	C-CELL	Mathew
☐	Press 96 Annual PM	Press #96	C-CELL	Mathew
☐	Press 96 5000 Cycle PM	Press #96	C-CELL	Peter Ka
☐	Press 94 Semi-annual PM	Press #94	C-CELL	Mathew
☐	Press 94 Quarterly PM	Press #94	C-CELL	Mathew
☐	Press 94 Annual PM	Press #94	C-CELL	Mathew
☐	Press 94 5000 Cycle PM	Press #94	C-CELL	Peter Ka
☐	Press 93 Semi-annual PM	Press #93	C-CELL	Mathew

Menu

Dashboard

Calendar

Work Orders

Assets

Parts & Purchase Orders

Safety Programs

Admin

Revere Plastics Systems, LLC.

CONFIGURATIONS

+ New

Delete

Import

Export

Task Books

☐	Name	Site	Description
☐	Cincinatti Milacron Annual PM		Annual PM for Cincinatti Milacron Machines
☐	Cincinatti Milacron Semi-Annual PM		Semi-Annual PM for Cincinatti Milacron Machines
☐	DEMAG Annual PM		Annual PM for DEMAG Machines
☐	DEMAG Monthly PM		Monthly PM for DEMAG Machines
☐	DEMAG Quarterly PM		Quarterly PM for DEMAG Machines
☐	DEMAG Semi-annual PM		Semi-annual PM for DEMAG Machines
☐	Monthly Split Lock		Monthly task list for Ram style Split lock Presses.
☐	Monthly Toggle		Monthly Toggle PM Tasks
☐	Nissei Annual PM		Annual PM Template for Nissei Machines
☐	Nissei Semi-Annual PM		Semi-Annual PM for Nissei Machines
☐	Quarterly IMM PM		Quarterly Press PM's.

- 124 Mold Machines x 4 Type PMs = 496 Press PMs to schedule: $412 / 496 = 90\%$ Complete
- Added Robots: $82/82 = 100\%$ Complete
- SEPRO Robot PM schedule: $53/53 = 100\%$ Complete

Created Updated Robot Database

Documented relevant info for all 82 robots

Cell (Location)	Press (Parent Asset)	Serial #	Model	Manufacturer	Date of Manufacturer
A	116	P16285	AXESS22 S3 TRANS	SEPRO	2010
A	305	R006199-01	Multi Inject 30	SEPRO	2018
A	405	R006199-04	Multi Inject 30	SEPRO	2018
A	216	R006199-08	Multi Inject 30	SEPRO	2018
B	169	R006199-05	Multi Inject 30	SEPRO	2018
B	1403	R006199-07	Multi Inject 30	SEPRO	2018
B	164	R006199-03	Multi Inject 30	SEPRO	2018
B	1301	R006199-02	Multi Inject 30	SEPRO	2018
C	150	P12775	S4	SEPRO	2006
D	53	P17906	AXESS	SEPRO	
E	70	R014351-01	SUCCESS 33 8U30	SEPRO	2022
E	89	R013682-01	SUCCESS 33 8U30	SEPRO	2021
F	8124	5578	CONAIR S 900 II	SEPRO	1999
F	84	P19504	SUCCESS 22	SEPRO	
J	1302	R006199-06	Multi Inject 30	SEPRO	2018
J	158	P20110	SR AXESS44 S3 TRANS	SEPRO	2013
J	8174	R001866-02	S7-45	SEPRO	2016
J	8175	R001688-01	S7-45	SEPRO	2016
K	165	P12776	CONAIR A3	SEPRO	2006
K	182	P12107	CONAIR S4	SEPRO	2005
K	1601	P20458	SUCCESS 33	SEPRO	
K	209	P12564	CONAIR A3	SEPRO	2005
K	219	P19528	SUCCESS 22	SEPRO	
K	1602			SEPRO	
K	154	P13366	CONAIR AZ-VISUAL	SEPRO	2005
K	160	P12106	CONAIR S4	SEPRO	2005
K	159	P13626	CONAIR SE3	SEPRO	2007
O	8173	R002774-01	S7-45	SEPRO	2017
O	11159	P12104	CONAIR S3	SEPRO	2005
O	11160	P12105	CONAIR S3	SEPRO	2005
O	1404	P12980	CONAIR S3	SEPRO	2006
O	218	P12979	CONAIR S3	SEPRO	2006
O	1401	P20786	SR AXESS44 S3 TRANS	SEPRO	2014
O	1402	P20787	SR AXESS44 S3 TRANS	SEPRO	2014
O	2347	P13234	AXESS30	SEPRO	2006
S	1501	P23840	SR 4050 HL S3 AXIAL	SEPRO	2015

> ... **Robots** SEPRO FANUC UR AUBO WITTMANN KAWAS

Automated Transition to Brightly

- Created programs to add:
 - Press PMs,
 - Robots
 - SEPRO Robot PMs
- Efficient, better organized implementation

```
*PMcheat.pyw - C:\Users\gbahner\AppData\Local\Programs\Python\Python313\PMcheat.py...
File Edit Format Run Options Window Help
presses = array.array('a', [UBEpresents, DEMAGpresses, TOYOpresents, TAYUpresents, VANDOR
UBEpresents = array.array('i', [1601, 1401, 1402, 218, 1404, 210, 211, 212, 213, 217])
DEMAGpresses = array.array('i', [166, 165, 1115])
TOYOpresents = array.array('i', [2352])
TAYUpresents = array.array('i', [2301])
VANDORNpresents = array.array('i', [2363, 2342])
JSWpresents = array.array('i', [214, 215])
CINCINNATipresents = array.array('i', [106, 107])
TOSHIBApresents = array.array('i', [216, 405])
NISSEIpresents = array.array('i', [55, 56, 41])
PMtime = array.array('S', ["Monthly", "Quarterly"])
pg.moveTo(955, 955)
pg.click()
for press in pr
    #More
    pg.moveTo(3, 3)
    pg.click(39)
    time.sleep(1)
    #Duplicate
    pg.moveTo(4, 4)
    pg.click()
    time.sleep(1)
    #Drag, delete
    pg.moveTo(3, 3)
    pg.click()
    pg.dragTo(5, 5)
    pg.press('b')
    pg.write("P")
    #Tabbing general
    pg.moveTo(1, 1)
    pg.click()
    #Meter
    pg.moveTo(2, 2)
    pg.click(22)
    time.sleep(1)

    pg.moveTo(9, 9)
    pg.click(92)
    time.sleep(1)

    for index, row in subdata.iterrows():
        press = row['Press (Parent Asset)']
        pg.moveTo(460, 245, 1, pg.easeOutQuad)
        pg.click(460, 245, duration=0.5)
        time.sleep(2)
        #Duplicate
        pg.moveTo(460, 295, 0.5, pg.easeInOutQuad)
        pg.click()
        time.sleep(10)
        #Drag, delete, type in new name
        pg.moveTo(428, 364, 1, pg.easeInBounce)
        pg.click()
        pg.dragTo(670, 369, duration=2, button='left')
        pg.press('backspace')
        pg.write("SEPRO Press " + str(press) + " PM")
        time.sleep(1)
        #Description
        pg.tripleClick(425, 897, duration=1) #box loca
        pg.press('backspace')
        pg.write("PM for SEPRO Robot associated with Press "
        #Tabbing general
        pg.moveTo(221, 317, 1, pg.easeInElastic)
        pg.click()
        #Meter
        pg.moveTo(295, 608, duration=1)
        pg.click(295, 608, duration=1)
        time.sleep(3.5)
        pg.click(960, 500, duration=1)
        time.sleep(4)
        pg.click(705, 916, duration=1) #filter
        time.sleep(2.5)
        pg.click(1038, 471, duration=1) #plus button
        time.sleep(1)

        if serial != "" and serial != "nan":
            pg.tripleClick(1240, 500, duration=1) #serial
            pg.press('backspace')
            pg.write(serial)
        else:
            pg.tripleClick(1240, 500, duration=1)

Robot PM.py - C:\Users\gbahner\AppData\Local\Programs\Python\Python313\Robot PM.py...
File Edit Format Run Options W
for index, row in subdata.iterrows():
    location = str(row['Cell'])
    press = row['Press (Parent Asset)']
    manufacturer = str(row['Manufacturer'])
    serial = str(row['Serial'])
    model = str(row['Model'])
    if isinstance(press, int):
        asset = str(manufacturer)
    else:
        asset = str(manufacturer)
    year = row['Date of Manufacture']
    if isinstance(year, int):
        year = '01/01/' + str(year)

    pg.click(480, 245, duration=1)
    pg.click(480, 290, duration=1)
    time.sleep(8)
    #while not(pixelMatchesColor):
    #    time.sleep(1)

    pg.moveTo(9, 9)
    pg.click(92)
    time.sleep(1)

    pg.tripleClick(540, 445, duration=1)
    pg.press('backspace')
    pg.write(asset)

    if isinstance(year, str):
        if year != prevYear:
            pg.tripleClick(540, 445, duration=1)
            pg.write(year)
    else:
        pg.tripleClick(540, 445, duration=1)
        pg.press('backspace')

    if serial != "" and serial != "nan":
        pg.tripleClick(1240, 500, duration=1) #serial
        pg.press('backspace')
        pg.write(serial)
    else:
        pg.tripleClick(1240, 500, duration=1)
```

Next Steps

- Connect IQMS with Brightly, connect asset with correct IQMS input
- Determine correct cycle amount for each type of PM of each asset
- Add manuals and pictures for presses/robots
- Add other vital equipment with associated manuals, pictures, and PM tasks

Smaller Projects

Scientific Molding

- Setup new process for Press 127 with Travis

Cavity Balance Data

Mold #

Date

1/0/00

*Shoot parts till 95% filled. Record each cavity part weight

Cavity #	Part Weight (gm)				
1					
2					
3					
4					
5					
6					
7					
8					
9					
10					
11					
12					
13					
14					
15					
16					

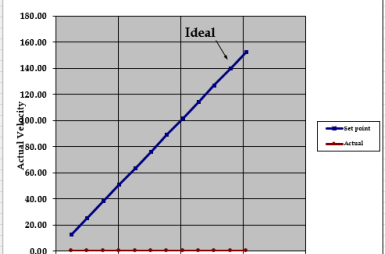
Max Part Wt - Min Part Wt
Max Part Wt

x 100

Max Part Wt. 0.000 gm
Min Part Wt. 0.000 gm
Ave. Part Wt. #DIV/0! gm

Shot #	Set Velocity in-mm/sec	Ft seconds	pressure psi	Ratio Ri	Actual Vel in-mm/sec	of set point	barrel used 25.65% is best
1	152.40	0.00	0	1.0	#####		
2	139.70	0.00	0	1.0	#####		
3	127.00	0.00	0	1.0	#####		
4	114.30	0.00	0	1.0	#####		
5	101.60	0.00	0	1.0	#####		
6	88.90	0.00	0	1.0	#####		
7	76.20	0.00	0	1.0	#####		
8	63.50	0.00	0	1.0	#####		
9	50.80	0.00	0	1.0	#####		
10	38.10	0.00	0	1.0	#####		
11	25.40	0.00	0	1.0	#####		
12	12.70	0.00	0	1.0	#####		

Actual vs Set Velocities



	A	B	C	D	E	F	G	H	I
1	Date	Machine & size				Intensification Ratio (Ri)			1.0
2	Mold #					Stroke, include decompression (inches or mm)			
3	Material & Melt Temp.					1st-2nd stage position transfer (inches or mm)			
4	Part					Stroke travel distance (inches or mm)			0.00
5		Mach. set	Fill Time	Hydraulic press.	Inten.	Plastic	Relative	Shear	Mach.
6		Velocity	Ft	at transfer (Ht)	Ratio	pressure psi	Viscosity	Rate	Actual Vel.
7	Shot #	in-mm/sec	seconds	psi	Ri	Ht x Ri	Ht x Ri x Ft	1/Ft	in-mm/sec
8	1	152.40			1.0	0			
9	2	139.70			1.0	0			
10	3	127.00			1.0	0			
11	4	114.30			1.0	0			
12	5	101.60			1.0	0			
13	6	88.90			1.0	0			
14	7	76.20			1.0	0			
15	8	63.50			1.0	0			
16	9	50.80			1.0	0			
17	10	38.10			1.0	0			
18	11	25.40			1.0	0			
19	12	12.70			1.0	0			

Machine Set-up Sheet (Toshiba - New)			
PAR1	DESCRIPTION	MATERIAL #	
MACHIN	MOLD/CAV #	Completed by:	
CLAMP SETUP			
LS 2D - (CLOSE POS.	.5MM	MM	
LS 2E - (CLOSE POS. R2	.5MM	MM	
LS 2A - (SLOWDOWN	.5MM	MM	
LS 2 - (MOLD TOUCH	.5MM	MM	
VO 1 - (CLOSE SPEED	.5X	X	
VO 2 - (CLOSE SPEED	.5X	X	
VO 3 - (CLOSE SPEED	.5X	X	
VO 5 - (CLOSE SLOW	.5X	X	
PCL - (MOLD PROTEC	.5X	X	
PCH - (TONNAGE)	.5X	X	
VO S2 - (OPEN SLOW	.5X	X	
VO 3 - (FAST OPEN 3	.5X	X	
VO 2 - (FAST OPEN 2	.5X	X	
VO 1 - (FAST OPEN 1	.5X	X	
VO S1 - (SLOW OPEN	.5X	X	
LS 3 - (OPEN LIMIT)	.5MM	MM	
LS 3B - (OPEN SLOW	.5MM	MM	
LS 3E - (OPEN FAST	.5MM	MM	
LS 3D - (OPEN FAST	.5MM	MM	
LS 3A - (START OPE	.5MM	MM	
LS 100	.5MM	MM	
LS 30	.5MM	MM	
THKMS	.5MM	MM	
LSX - LS 3M			
EJECT CONTROL			
LS 31A - (FWD POS 1)	.5X	MM	
LS 31 - (FWD LIMIT)	.5	MM	
VE 1 - (FWD SPEED	.5X	X	
VE 2 - (FWD SPEED 2)	.5X	X	
VR - (RETRACT SPEED)	.5X	X	
LS 32 - (RETRACT LIMIT)	.5MM	MM	
EJECT COUNTS			
EJECT MODE OFF			
TEMPERATURE CONTROL			
SETTINGS			
HEM - (NO2)	+/- 10	ACTUALS	DEG
H1 - (FROM	+/- 10		DEG
H2 - (F. CE	+/- 10		DEG
H3 - (R CEI	+/- 10		DEG
H4 - (REAR	+/- 10		DEG
TIMER SET-UP			
INJECT HIGH	.5SEC	SEC	
COOLING	.5SEC	SEC	
INT - (OPEN DWELL)	.5SEC	SEC	
TR H3 - (HOLD TIME)	.5SEC	SEC	
TR H2 - (PACK TIME 2)	.5SEC	SEC	
TR H1 - (PACK TIME 1)	.5SEC	SEC	
TRVH	.5SEC	SEC	
INJECTION			
VI 1 - (INJECT SPEED 1)	.5X	X	
VI 2 - (INJECT SPEED 2)	.5X	X	
VI 3 - (INJECT SPEED 3)	.5X	X	
VI 4 - (INJECT SPEED 4)	.5X	X	
VI 5 - (INJECT SPEED 5)	.5X	X	
VH 1 - (PACK SPEED)	.5X	X	
VH 2 - (HOLD SPEED)	.5X	X	
LS 10 - (DECOMPRESS)	.5MM	MM	
LS 5 - (SHOT SIZE)	.5MM	MM	
LS 4A - (INJECT POS 1)	.5MM	MM	
LS 4B - (INJECT POS 2)	.5MM	MM	
LS 4C - (INJECT POS 3)	.5MM	MM	
LS 4D	.5MM	MM	
LS 4 - (TRANSFER)	.5MM	MM	
HYD - (PSI 2-FER)	.5PSI	PSI	
PI 1 - (INJECT HIGH PS	.5X	X	
PH 1 - (PACK PSI 1)	.5X	X	
PH 2 - (PACK PSI 2)	.5X	X	
PH 3 - (HOLD PSI 1)	.5X	X	
PH 4 - (HOLD PSI 2)	.5X	X	
SRN - (SCREW RPM'S)	.5X	X	
BP - (BACK PRESSURE)	.5X	X	
AUXILIARY EQUIPMENT			
CHILLER TEMPERATURE		DEG	
THERMOLATOR TEMPERATURE		DEG	
TOWER HOSES	LIVE	DEAD	
CHILLER HOSES	LIVE	DEAD	
THERMOLATOR HOSES	LIVE	DEAD	
NOZZEL TIP:		TYPE	

Quality Assurance Device

Used Lathe in Tool Room

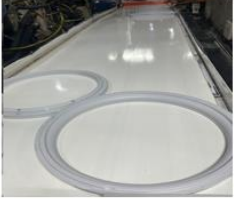
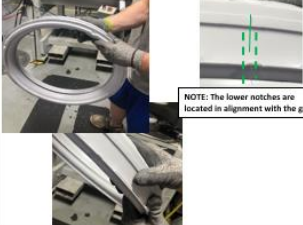
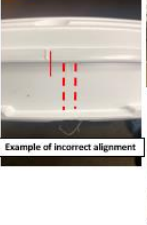
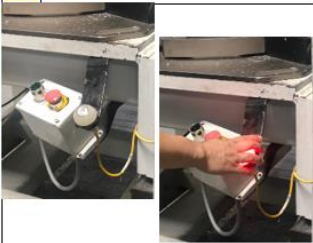
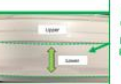
Helps ensure proper alignment of nozzle

- Used as intermediate corrective action
- Easily iterable: made from part itself



VA-08 Standard Work Procedure

- Created new operator work guide for VA-08 process using new template

Operation: Weld Process			
Part Name: Balance Ring Assembly VA08 SS		Station:	Customer: Whirlpool Clide
Part No: V1111375	Revision: 1	Date: 8/7/2025	Sheet: 1 of 3
1 Retrieve Upper (W1113752) and Lower (W1113754) from 	2 Inspect Upper and Low for Defects 	3 Align notches (shown in picture), then place into the welder  NOTE: The lower notches are located in alignment with the gate.	4 Adjust and ensure correct alignment when inside the welder 
How:	How:	How:	How:
5 After correctly placing part in welder, press finger sensor switch to start weld machine cycle. 	6 After machine cycle finishes, remove part and check inside edge for snap points with cheesecloth. Check the outer weld for strings/drool 	7 Verify welded part has correct notch alignment (single mark within 2 lower marks). 	8 Move to leak testing guide on page 2. 
How:	How:	How:	How:
A) WEIGHT - Weight is out of specification B) PLUG - Part is missing plug C) LEAK - The plug failed leak test.			
How:	How:	How:	How:
8 Place any defective parts from the process in the defect area and record the defect 			
How:	How:	How:	How:
Hand-written date MUST be located between the 2 green lines, on the Outer Diameter 			
How:	How:	How:	How:
Need Fail Pic 			
How:	How:	How:	How:

Fabrication of Non-Magnetic Nozzle

Used Milling Machine and Lathe in Tool Room

No magnetic effect, explored possibility of reduced glue flow rate



Other Smaller Projects

- Press/Quality study with Conor – Whirlpool approved second sample
- Setup new drum roller, helped reprogram FANUC for drum roller with Ciara
- Fusion initial leak tester machine, carousel seal repair with Ciara

Final Takeaways

- Managing expectations, accountability and responsibility
- Importance of open-ended projects and freedom
- Thanks Travis



**Revere Plastics
Systems, LLC.**